

Ch – 4 Determinants

Determinants of a square matrix (up to 3x3 matrices), minors, co-factors, and applications of determinants in finding the area of a triangle. Adjoint and inverse of a square matrix. Consistency, inconsistency, and number of solutions of system of linear equations by example, solving system of linear equations in two or three variables (having unique solution) using inverse of matrix.

Determinants

Every square matrix can be associated to an expression or a number which is known as its determinant. If A is a square matrix then its determinant is denoted by $\det A$ or $|A|$.

Note:

- (i) Only square matrices have determinants. The determinant of non-square matrices is not defined.
- (ii) A matrix is an arrangement of numbers and hence it has no fixed value while each determinant has a fixed value.
- (iii) A determinant having n rows and having n columns is known as a determinant of order n.

Value of a determinant

Determinant of a matrix of order one

Let $A = [a_{11}]$ be a square matrix of order 1 then $|A| = |a_{11}|$ and the value of determinant is the number itself. i.e. $|a_{11}| = a_{11}$

Remark: A determinant of order 1 should not be confused with the absolute value of the number a_{11} .

As for example: (i) If $A = [5]$, then $|5| = 5$

(ii) If $A = [-2]$, then $|-2| = -2$

Determinant of a matrix of order 2

Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ be a square matrix of order 2,

then $\det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$ and its value is $a_{11} \cdot a_{22} - a_{21} \cdot a_{12}$

Note: The numbers $a_{11}, a_{12}, a_{21}, a_{22}$ are called the elements of the determinant.

As for example: (i) If $A = \begin{bmatrix} 3 & 2 \\ 4 & 6 \end{bmatrix}$, then $|A| = 3 \times 6 - 4 \times 2 = 10$

(ii) $\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} = \cos\theta \cdot \cos\theta - \sin\theta \cdot (-\sin\theta) = \cos^2\theta + \sin^2\theta = 1$

Determinant of a matrix of order 3

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ be a square matrix of order 3,

then $\det A = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

$$= a_{11}(a_{22} \cdot a_{33} - a_{23} \cdot a_{32}) - a_{12}(a_{21} \cdot a_{33} - a_{23} \cdot a_{31}) + a_{13}(a_{21} \cdot a_{32} - a_{22} \cdot a_{31})$$

As for example: (i) $\begin{vmatrix} 3 & -2 & 5 \\ 1 & 2 & -1 \\ 0 & 4 & 7 \end{vmatrix} = 3 \begin{vmatrix} 2 & -1 \\ 4 & 7 \end{vmatrix} - (-2) \begin{vmatrix} 1 & -1 \\ 0 & 7 \end{vmatrix} + 5 \begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix} = 3.18 + 27 + 5.4 = 88$

(ii) $\begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix} = 3(0-5) - (-1)(0+3) + (-2)(0-0) = -15 + 3 + 0 = -12$

IMPORTANT POINTS TO REMEMBER

1. Only square matrices have determinants.
2. In case of matrices, we take out any common factor from each element of matrix, while in the case of determinants we can take out common factor from any one row or any one column of the determinant.
3. If area is given, then we take both positive and negative values of the determinant for calculation.
4. If we want to prove that three points are collinear, we show that the area formed by these three points is equal to zero.
5. A square matrix of order n is invertible if it is non-singular.
6. If A is an invertible matrix of order n , then $A^{-1} = \frac{1}{|A|} \text{adj } A$ where $|A| \neq 0$
7. The inverse of an invertible symmetric matrix is symmetric.
8. If a system of equations has one or more solutions, then it is said to be a consistent system of equations otherwise it is in-consistent.
9. A system of linear equations may or may not be consistent.
10. A consistent system may or may not have a unique solution.
11. Every invertible matrix possesses a unique inverse.
12. If A' is transpose of a square matrix A then $|A'| = |A|$

As for example: If $A = \begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix}$, then $|A| = 2.4 - 5.3 = 8 - 15 = -7$

Now $A' = \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}$, then $|A'| = 2.4 - 3.5 = 8 - 15 = -7$

13. If $A = [a_{ij}]_{n \times n}$ then $|kA| = k^n |A|$

As for example:

If $A = [a_{ij}]_{3 \times 3}$, then $|kA| = k^3 |A|$

14. If A is a non-singular matrix of order n , then $|\text{adj } A| = |A|^{n-1}$
15. If A is a non-singular matrix of order n , then $|A^{-1}| = \frac{1}{|A|}$
16. If A and B are non-singular matrices of same order, then $|AB| = |A||B|$
and hence $|A \cdot \text{adj } A| = |A|^n$

Illustrations

1. If A is square matrix of order 3 with value of its determinant 4,
then $|3A| = 3^3 |A| = 27 \times 4 = 108$
2. If A is a square matrix order 3 such that $|A| = 4$, and $|KA| = 500$, find K .
Solution: $\because |KA| = 500$
 $\Rightarrow k^3 |A| = 500$
 $\Rightarrow k^3 \cdot 4 = 500$
 $\Rightarrow k^3 = 500/4 = 125$
 $\Rightarrow k = 5$
3. Let A be a square matrix of order 3×3 , write the value of $|2A|$ where $|A| = 4$

Solution: $\because |2A| = 2^3 \cdot |A| = 8 \times 4 = 32$

4. Given matrix A of order 3x3, find the value of k such that $|2A| = K|A|$

Solution: $\because |2A| = 2^3 \cdot |A| = K|A|$

$$\Rightarrow K = 8$$

Area of a triangle

Area of a triangle with vertices (x_1, y_1) , (x_2, y_2) , (x_3, y_3) is given by

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

- Q. Find the value of K, if area of a triangle is 4 square unit when its vertices are $(k, 0)$, $(4, 0)$, $(0, 2)$.

(Ans: $k=0$ or 8)

Adjoint of a matrix

Let $A = [a_{ij}]_{n \times n}$ be a square matrix of order n and let A_{ij} denote the cofactor of a_{ij} in $|A|$. Then the adjoint of A denoted by $\text{adj } A$ and is defined by $\text{adj } A = [A_{ji}]_{n \times n}$. Thus, $\text{adj } A$ is the transpose of the matrix of the corresponding cofactors of elements of $|A|$.

As for example: Let $A = \begin{bmatrix} 2 & -3 \\ 4 & 7 \end{bmatrix}$, then

$$M_{11} = |7| = 7; \quad M_{12} = |4| = 4; \quad M_{21} = |-3| = -3; \quad M_{22} = |2| = 2$$

$$A_{11} = (-1)^{1+1} M_{11} = |7| = 7; \quad A_{12} = (-1)^{1+2} M_{12} = -|4| = -4$$

$$A_{21} = (-1)^{2+1} M_{21} = -|-3| = -(-3) = 3 \quad A_{22} = (-1)^{2+2} M_{22} = |2| = 2$$

$$\text{adj } A = \text{transpose of } \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ -4 & 2 \end{bmatrix}$$

Q1. Find adjoint of a matrix $\begin{bmatrix} 3 & 1 \\ -5 & 4 \end{bmatrix}$

$$A_{11} = (-1)^{1+1} M_{11} = |4| = 4; \quad A_{12} = (-1)^{1+2} M_{12} = -|-5| = 5$$

$$A_{21} = (-1)^{2+1} M_{21} = -|1| = -1 \quad A_{22} = (-1)^{2+2} M_{22} = |3| = 3$$

$$\therefore \text{adj } A = \begin{bmatrix} 4 & -1 \\ 5 & 3 \end{bmatrix}$$